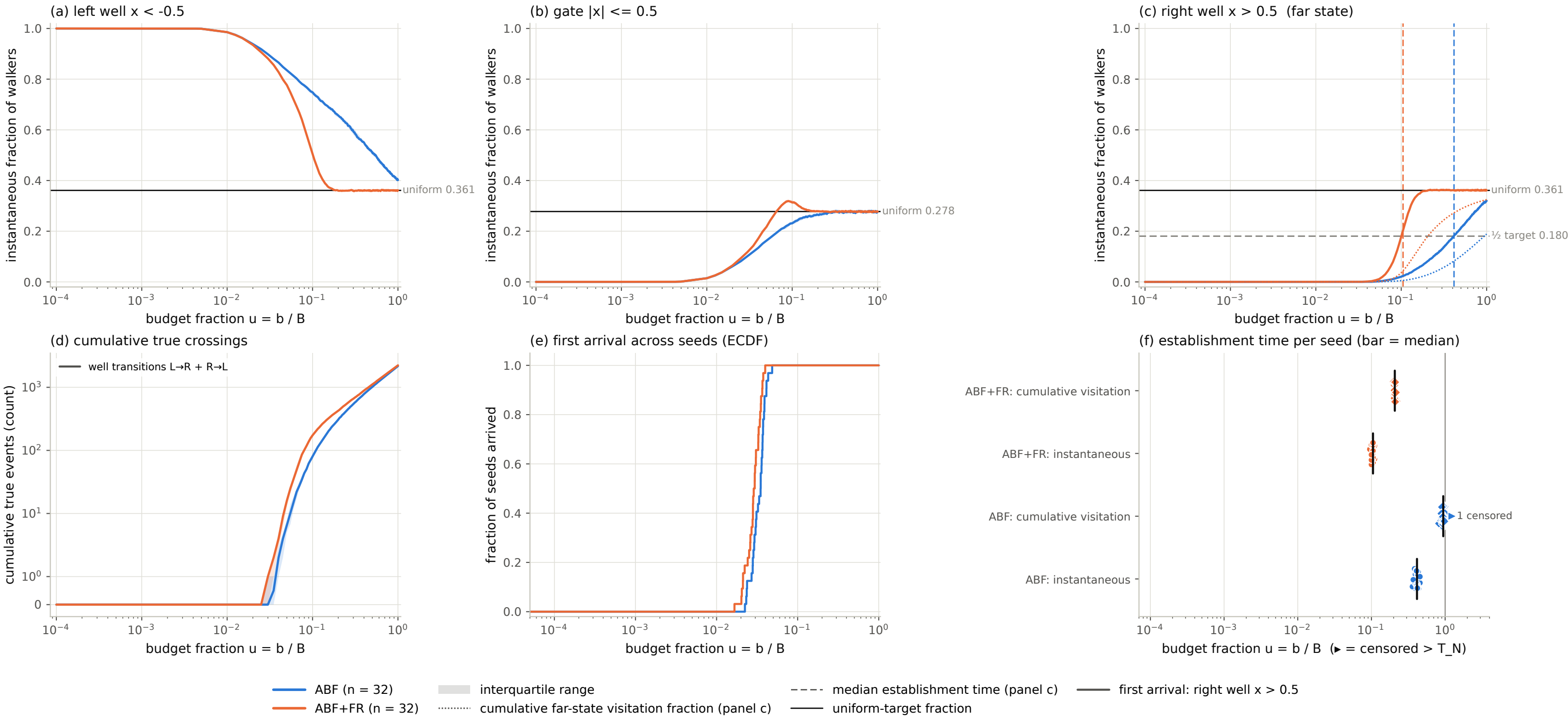


Gateway family conditional_relaxation/lam1 · N = 2048 · T_N = 40 t.u.

n_steps = 1,600,000 · h = 2.5e-05 · B = N n_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Establishment and crossings vs budget fraction u



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Establishment (frozen): first save after which the instantaneous far-state fraction (right well $x > 0.5$) stays $\geq \frac{1}{2}$ of its uniform target 0.361; the cumulative-visitation version uses C_all (far-state bins) against $\frac{1}{2}$ of their bin share. True events never count a copy (clones inherit labels).

Null calibration at N = 2048: an EXACTLY uniform population (far count $\sim \text{Bin}(N, 0.361)$ per save) fails the instantaneous criterion at a save with $p = 1.74\text{e-}72$ and passes every save of the second half with $P = 1.000$.